Review report on MATCOM-D-26-01427

Summary:

This manuscript applies the Parker–Sochacki method to the integration of charged particle motion in magnetic fields and compares its performance against several Runge–Kutta methods. While the numerical results appear technically correct and the presentation is clear, the work lacks substantive mathematical improvements, sufficient novelty, and meaningful algorithmic or theoretical contribution to warrant publication in a decent journal.

Major concerns:

(1) Lack of algorithmic novelty. The PS method is not new; it has been established in the numerical analysis literature for decades. The authors introduce no new convergence analysis, no adaptive strategy, and no theoretical extension.

(2) Benchmark comparisons are insufficient. The comparison against RK4, RK45, and RKG is too narrow. More competitive modern integrators (e.g., high-order implicit/explicit Runge–Kutta methods, multistep methods, or other series-expansion techniques) are not considered, which weakens the claimed performance advantages.

Recommendation: Reject. The authors may consider submitting to a more application-oriented journal.